

Dimensionality reduction methods

Introduction lecture and lecture 2

Course for Constructor University
JetBrains Academy



March-May 2026

Presented by

Vladimir Kukushkin

Liubov Tupikina

Outline

Lecture 1 (Vladimir). PCA, MDS, Local MDS, ISOMAP

Lecture 2 (Liubov). TSNE non-linear methods

Lecture 3 (Vladimir). SVD-decomposition, collaborative filtering, RecSys

Lecture 4 (Liubov). *2vec, autoencoders

Lecture 5 (Liubov). UMAP

Lecture 6 (Vladimir). Markov model, HMM, applications in user behavior analysis

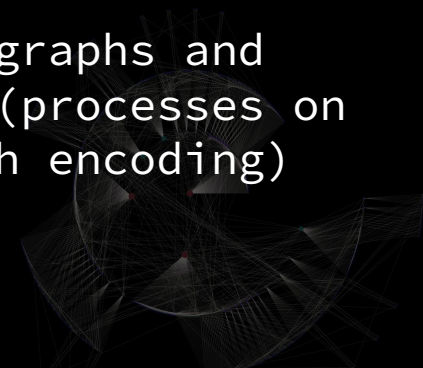
About us

Liubov

Researcher MIRAI, MIPT,
Nokia Bell Labs, France

Explainable AI (geometry of
embeddings, analysis of
higher-order data)

Geometry of graphs and
hypergraphs (processes on
graphs, graph encoding)



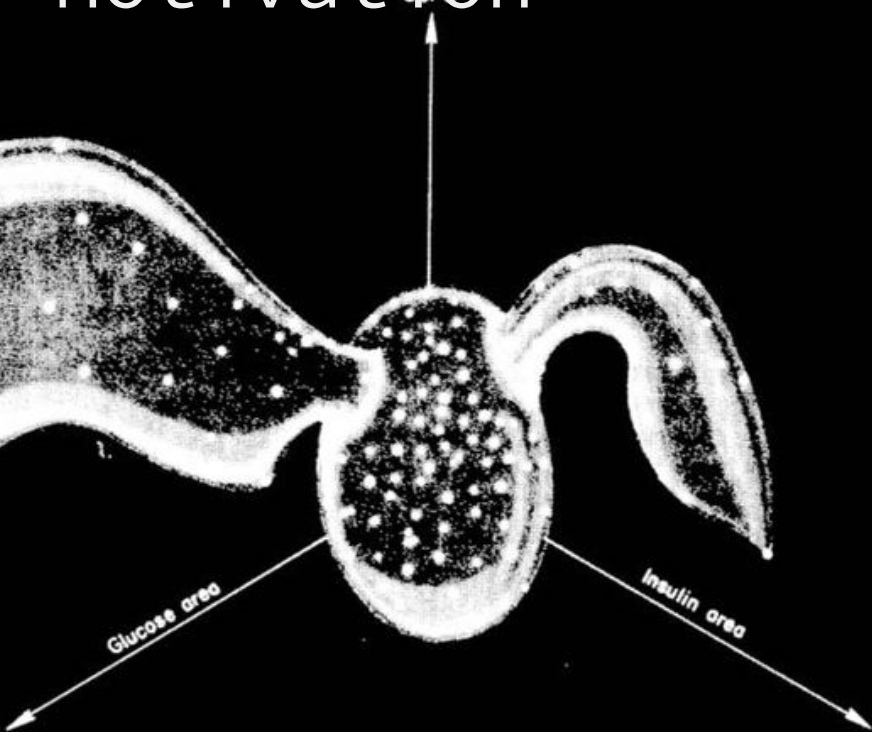
Vladimir

Data Scientist, ex.
Perplexity, ex. Yandex

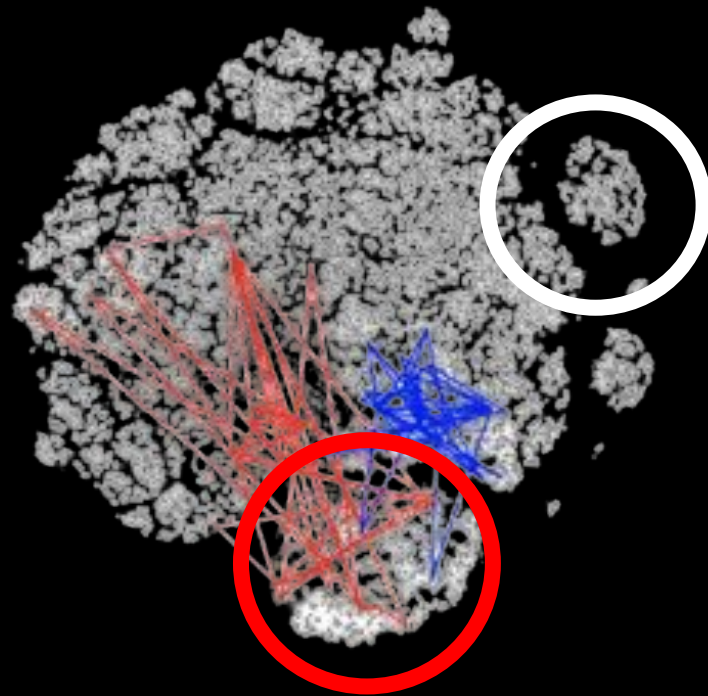
Quantitative UX research,
applications in product
analytics

Markov models, sequential
pattern mining, user behavior
embeddings and visualisations

Motivation



Textures of diabetes data

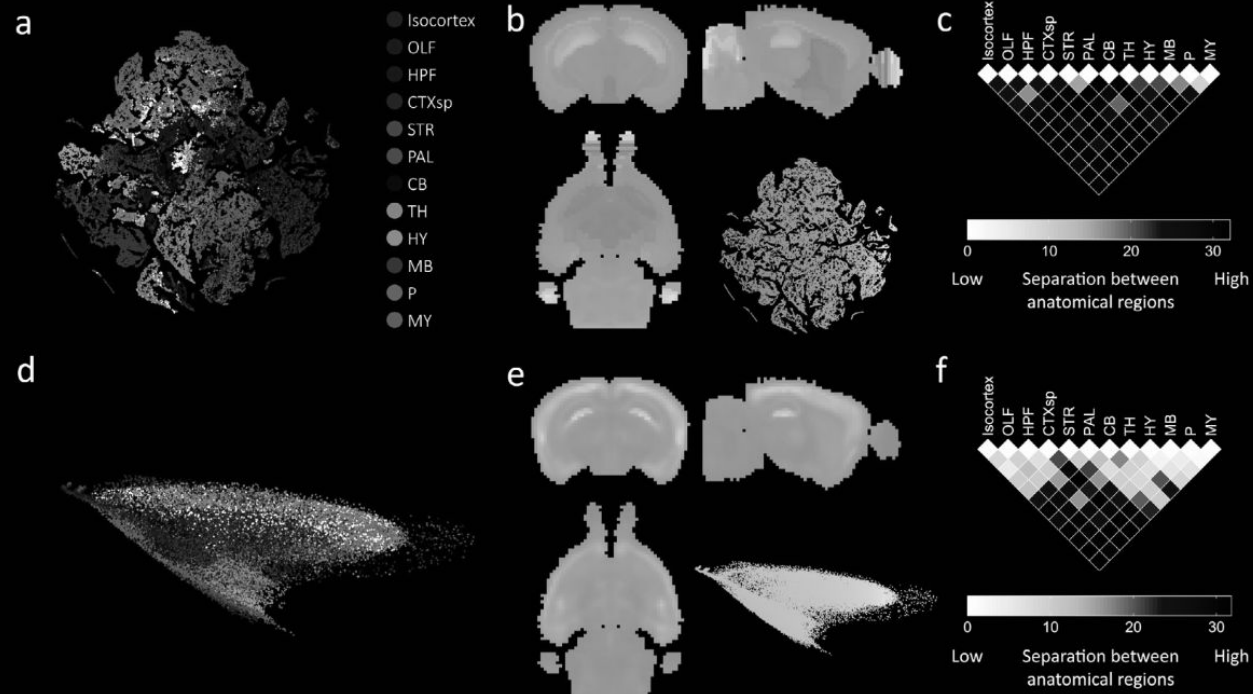


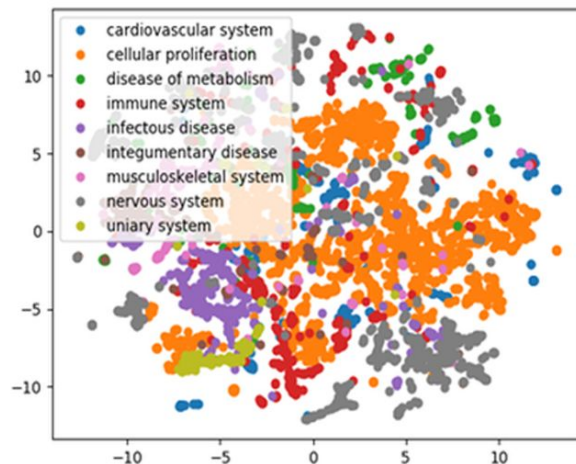
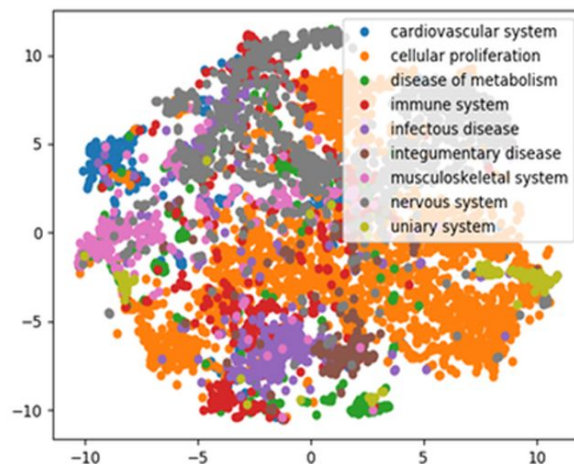
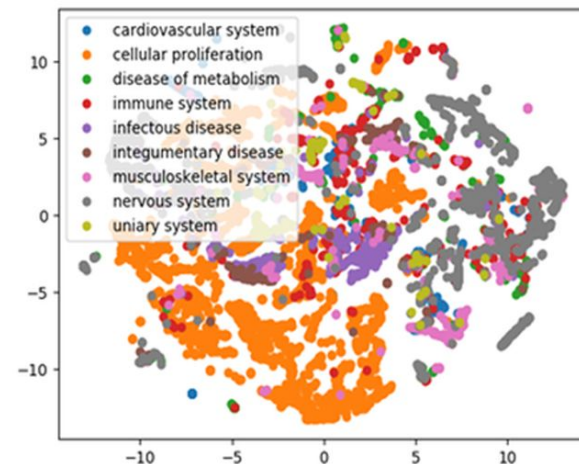
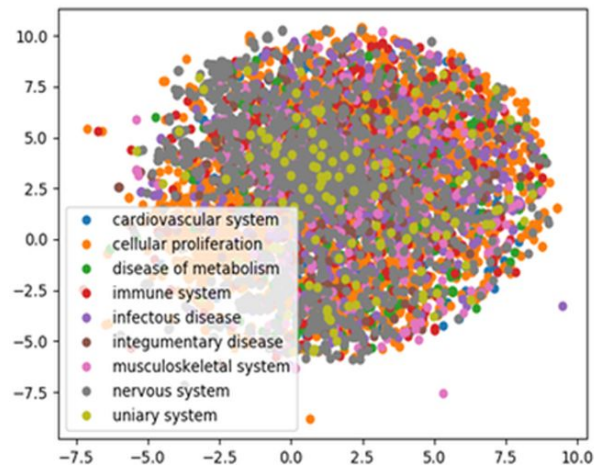
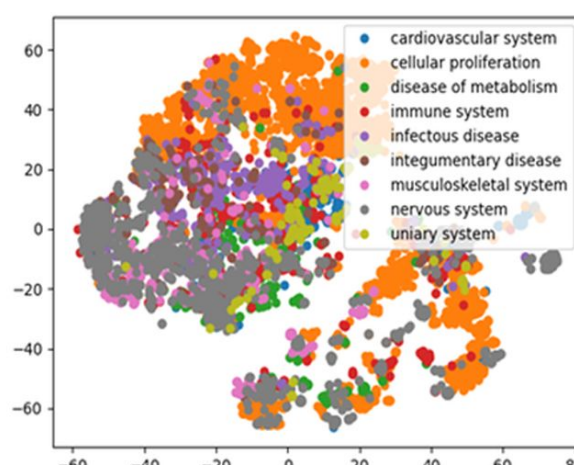
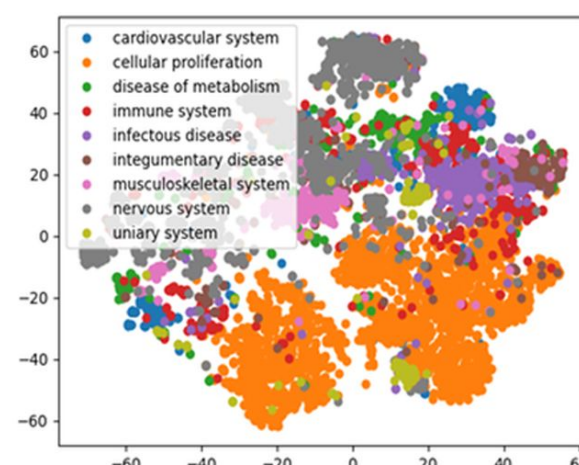
Walks in scientific articles data

Personal projects assignment

A. Mahfouz et al. / Methods 73 (2015) 79–89

83



A. Walking_RDF/OWL**B. TransE embeddings****C. Poincare embeddings****D. Rescal embeddings****E. SimpleE embeddings****F. R-GCN embeddings**

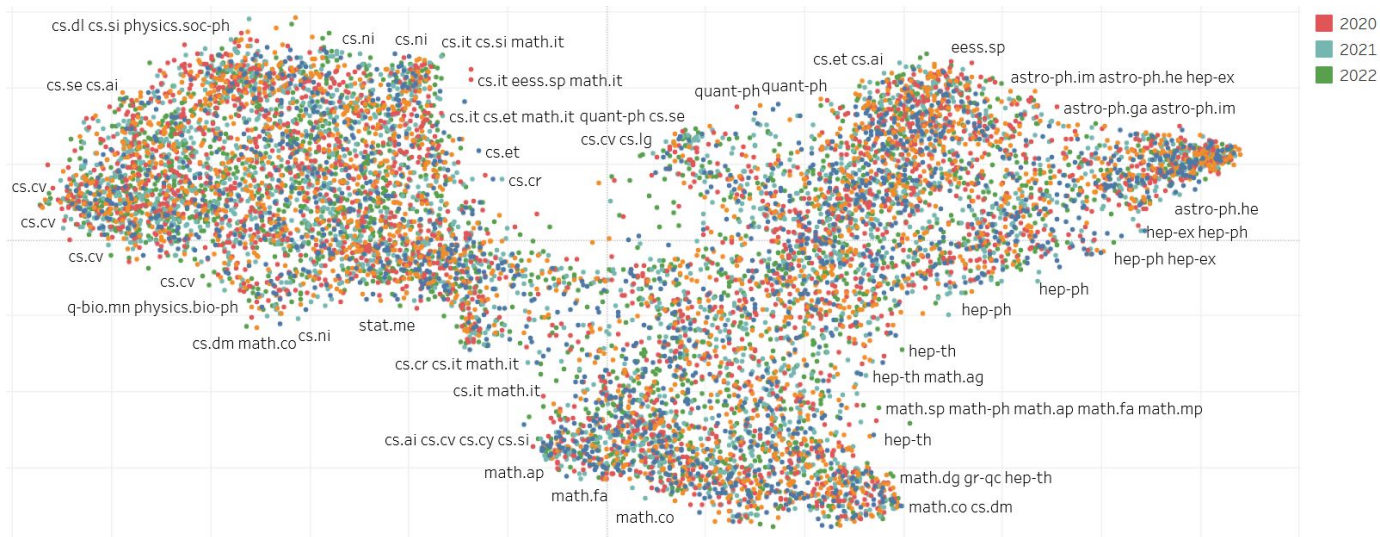
Personal projects assignment

- Take dataset from the suggested ones, choose the hypothesis

Example datasets are shown in [Drive of the course](#)

- Select and apply several embedding methods
- Compare the obtained results, quantitatively demonstrate and explain which methods express the data related question in better way
- Create and submit your project to [github](#) of the course here

Submit your personal project to the table.



Outline of coming lectures

Lecture 2 (Liubov) – t-SNE

- **t-SNE**
Technique that maps high-dimensional data to 2D/3D while preserving local similarities
Converts distances to probabilities and minimizes KL divergence
- ISOMAP: nonlinear method preserving geodesic distances along the data manifold

Lecture 4 (Liubov) – *2vec & Autoencoders

- **Word2vec / Item2vec**
Learns dense embeddings of discrete objects (words, items)
Similar objects have nearby vectors in embedding space
Neural architectures: skip-gram, CBOW
Applications: NLP, recommendations, graph embeddings
- **Autoencoders**
Unsupervised neural networks that learn efficient lower-dimensional representations by reconstructing input
Encoder-decoder structure with bottleneck
Variants: denoising, sparse, variational (VAE)
Applications: anomaly detection, denoising, generative modeling

Outline of lecture

Part 1:

Theory of tSNE method (Overleaf)

Illustration of the embedding tSNE method

Part 2:

Practice and coding: comparison of embedding methods

Applications to embedding framework on the data source examples

Main ideas and applications of tSNE method

Overleaf part

1. tSNE method notebook [link](#)
2. Advanced tSNE method from Fit-SNE et al. [link](#)
3. (Optional) Comparison of linear dimensionality reduction method [link](#)

Data for personal projects [link](#)

Contrastive learning and tSNE on large datasets

[Youtube lecture](#)

Personal projects ideas

Data for personal projects [link](#)

Check ideas from

<http://setosa.io/ev/principal-component-analysis/>

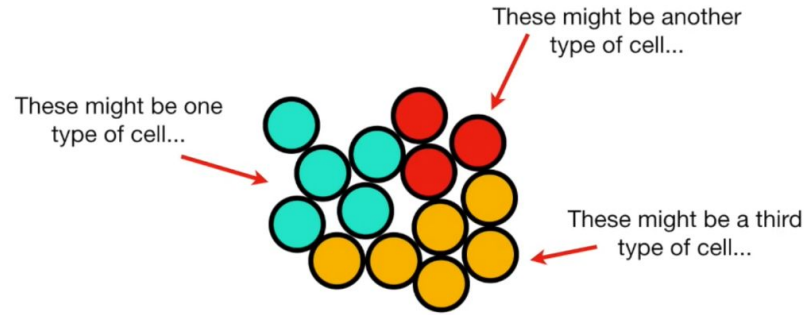
<https://distill.pub/2016/misread-tsne/>

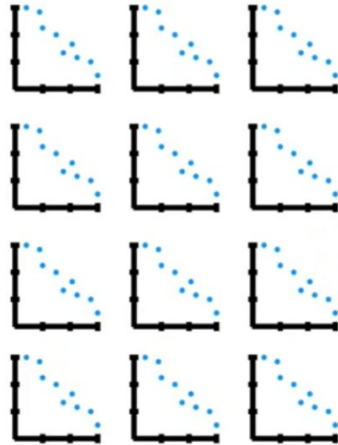
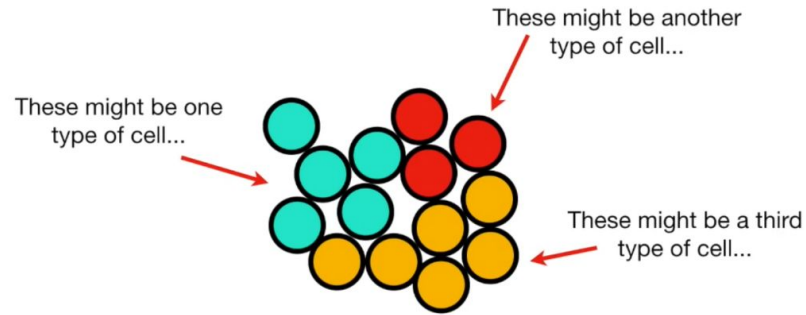
Extending Embedding Projector: Interactive Visualization and
Interpretation of Embeddings

[https://opensource.googleblog.com/2016/12/open-sourcing-
Embedding-projector-tool.html](https://opensource.googleblog.com/2016/12/open-sourcing-embedding-projector-tool.html)

<http://projector.tensorflow.org/>

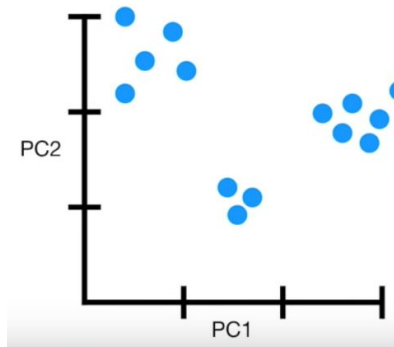
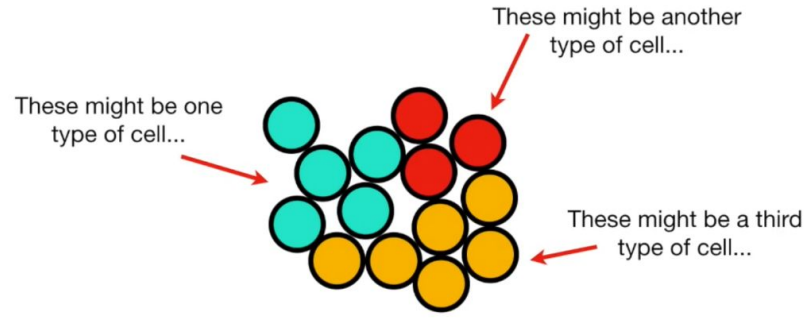
Recalling lecture 1





Draw tons and tons of 2 cell plots and try to make sense of them all?

	Cell1	Cell2	Cell3	Cell4	...
Gene1	3	0.25	2.8	0.1	...
Gene2	2.9	0.8	2.2	1.8	...
Gene3	2.2	1	1.5	3.2	...
Gene4	2	1.4	2	0.3	...
Gene5	1.3	1.6	1.6	0	...
Gene6	1.5	2	2.1	3	...
Gene7	1.1	2.2	1.2	2.8	...



	Cell1	Cell2	Cell3	Cell4	...
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Gene7	1.1	2.2	1.2	2.8	...

Recalling lecture 1

Why PCA or MDS may not be enough as methods for embeddings?

MDS — Idea

- ▶ Often we only care about preserving **pairwise distances**, not the absolute coordinates.
- ▶ MDS finds N points $z_1, \dots, z_N \in \mathbb{R}^q$ minimising the **stress function**:

$$S_M(z_1, \dots, z_N) = \sum_{i < i'} (d_{ii'} - \|z_i - z_{i'}\|)^2 \quad (3)$$

where $d_{ii'}$ are the original pairwise distances.

- ▶ MDS is a **non-linear** method (despite a linear-algebra solution for classical MDS).

Recalling lecture 1

What are main differences between MDS, local MDS and PCA?

MDS — Idea

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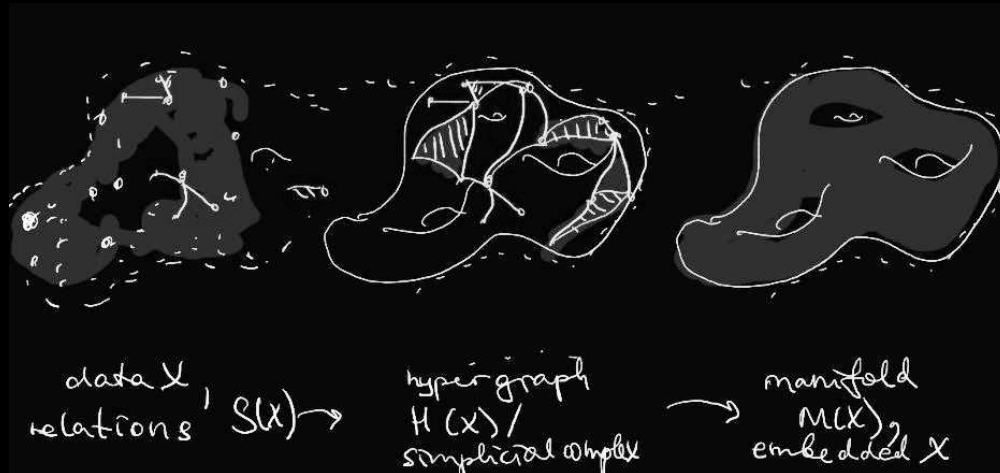
Recalling lecture 1

Why PCA Is Not Enough?

- Linear method
- Maximizes variance
- Preserves global Euclidean structure

Limitation:

- Cannot preserve nonlinear (data) manifolds
- Distances in high-D lose meaning
- Clusters often overlap after projection

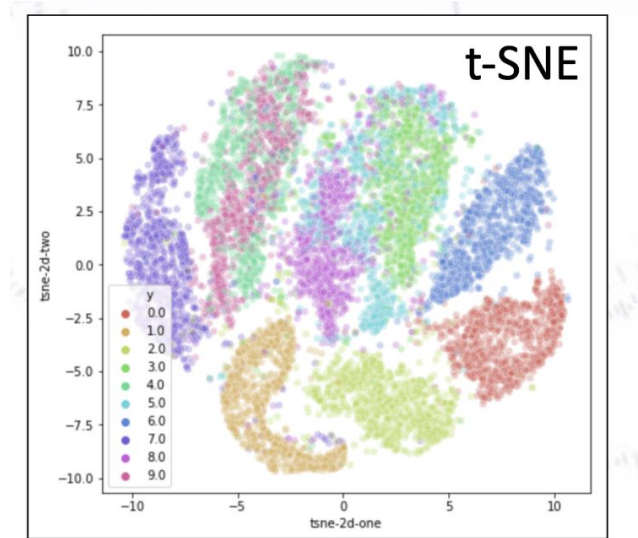


Other algorithms alternatives

<i>Techniques</i>	<i>Goal and Strength</i>	<i>Weaknesses</i>	<i>Computational complexity</i>
PCA	Maximize the variance with low noise sensitivity	Limited to linear projection	$O(d^2n + n^3)$
LDA	Maximize class separation	Suffer from class singularity issue	$O(d^2n)$, $n > d$; $O(d^3)$, $d > n$
SVD	Minimum construction error	Not suitable for nonlinear data	$O(d^2n + n^3)$
t-SNE	Preserve local structure	Provide only 2 or 3 features	$O(n^2)$
ICA	Maximize statistical independence	Require high training time	$O[2di(d+1)n]$

Other algorithms alternatives

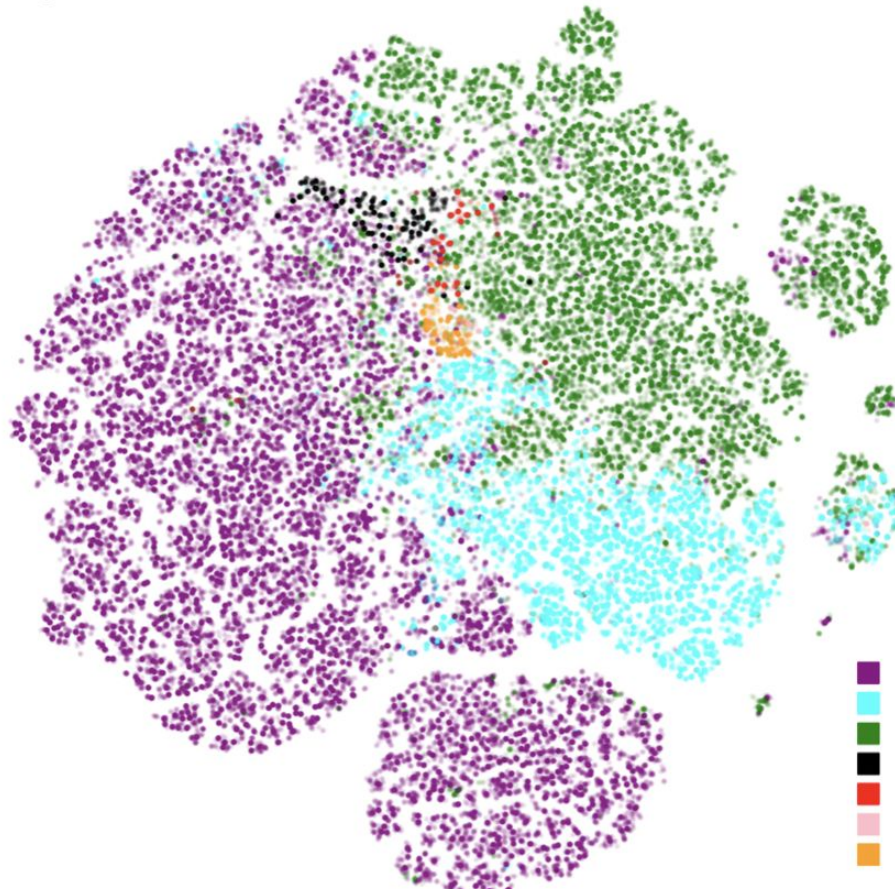
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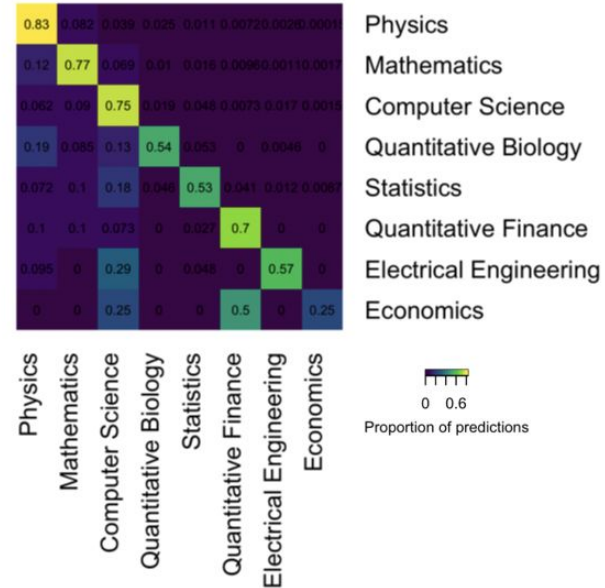
Source: Towards data science (PCA and t-SNE)

tSNE algorithm output

a

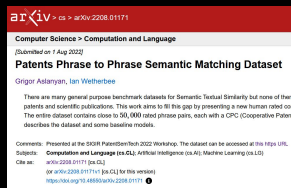
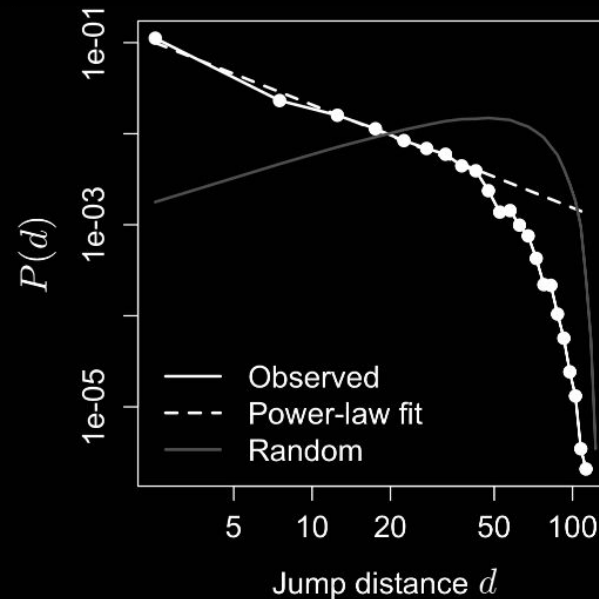
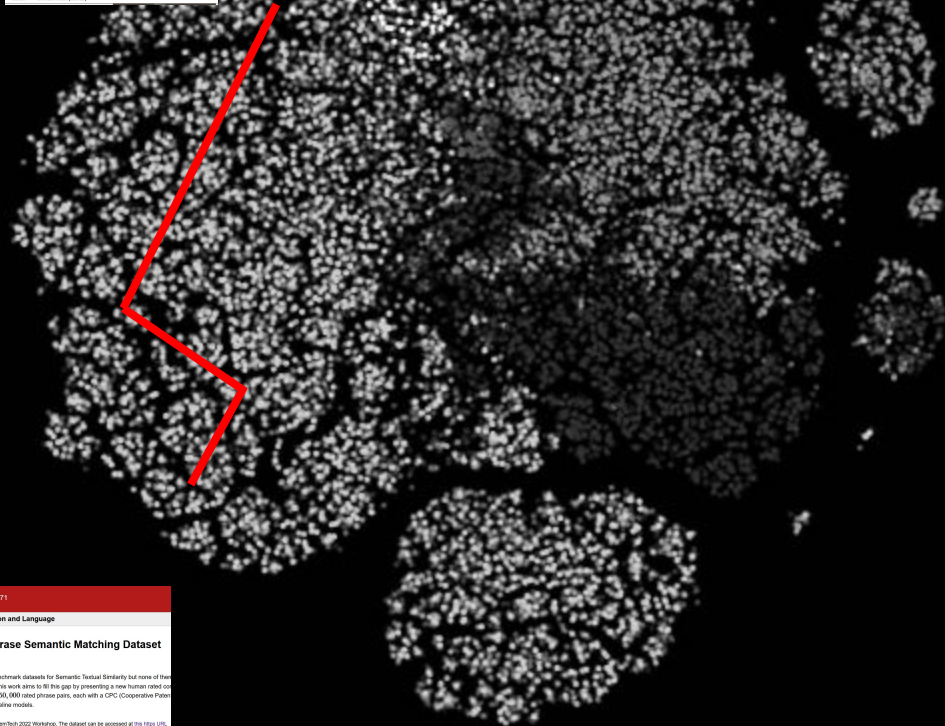


b

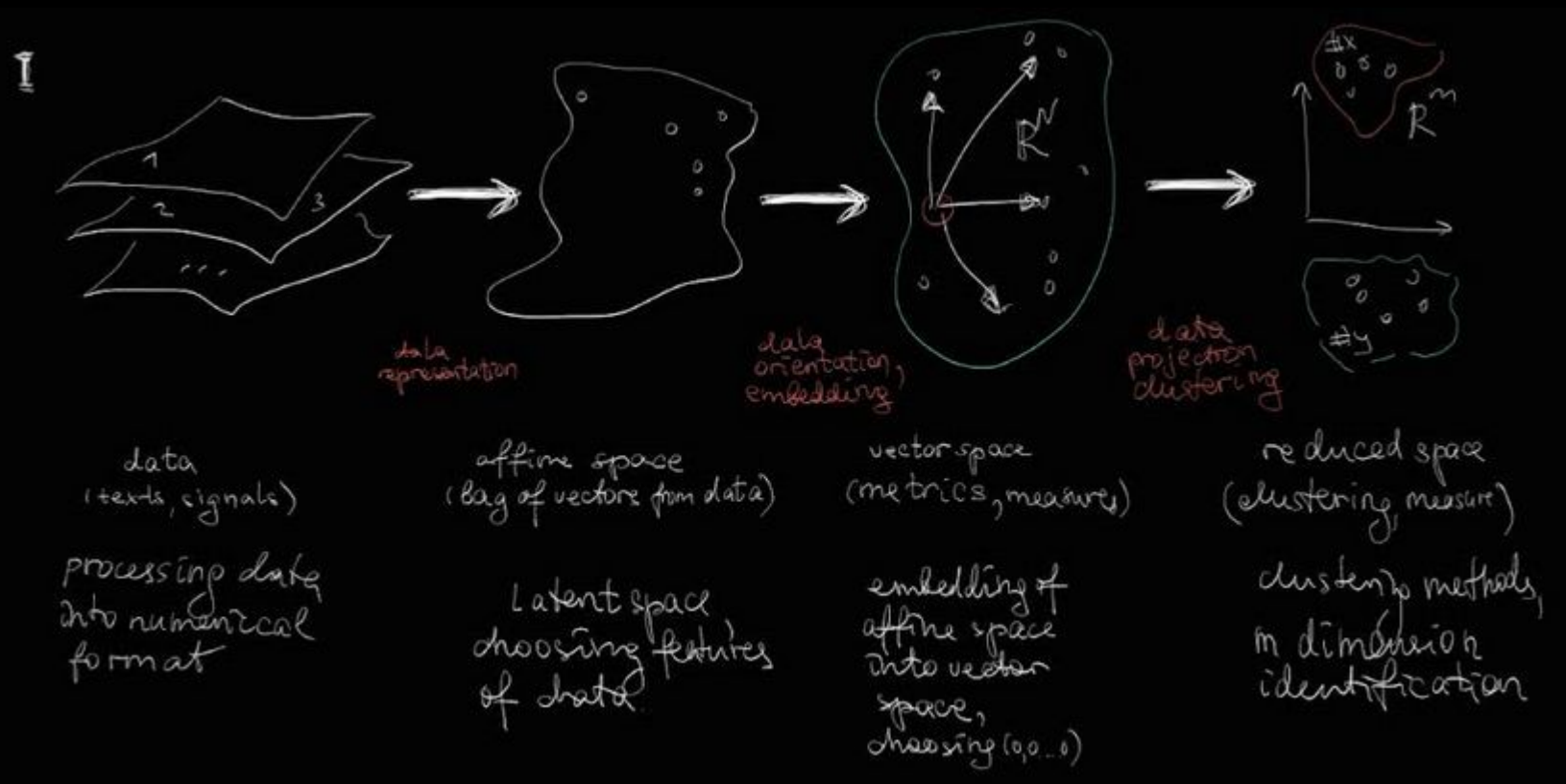


- Physics
- Mathematics
- Computer Science
- Quantitative Biology
- Statistics
- Economics
- Quantitative Finance

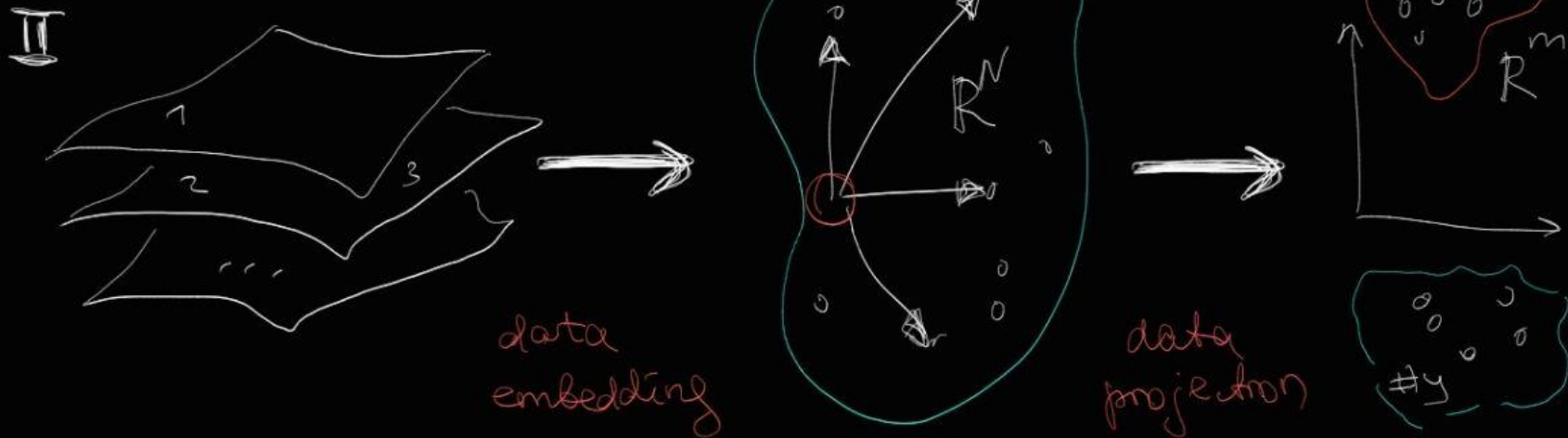
Singh, [...], [Santolini+](#) PLOS ONE 2022
Singh, [...], [Santolini+](#) EPJ (in rev) 2023



Main idea of embeddings

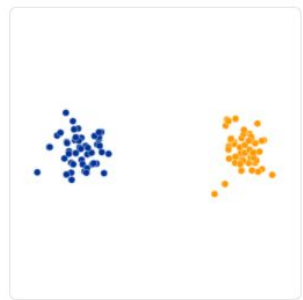


Main idea of embeddings for textual data

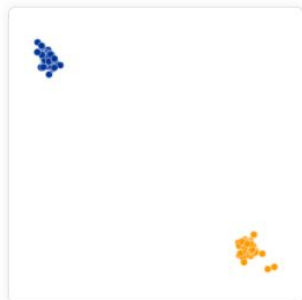


How to use tSNE effectively?

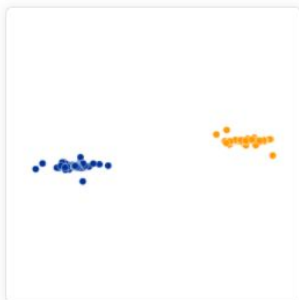
Parameters choice



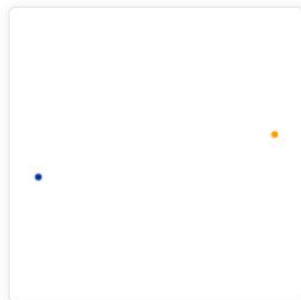
Original



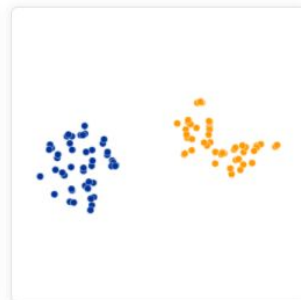
Perplexity: 30
Step: 10



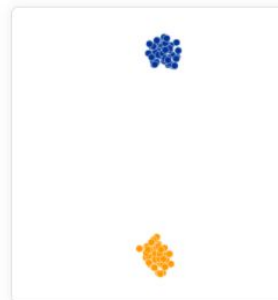
Perplexity: 30
Step: 20



Perplexity: 30
Step: 60



Perplexity: 30
Step: 120

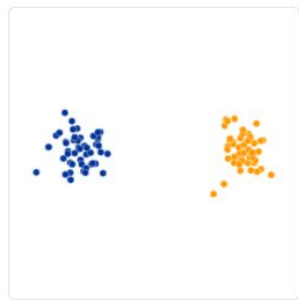


Perplexity: 30
Step: 1,000

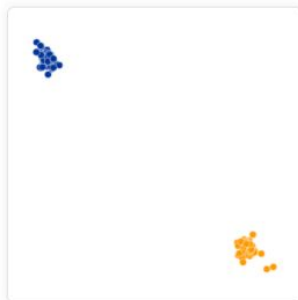
<https://distill.pub/2016/misread-tsne/>

How to use tSNE effectively?

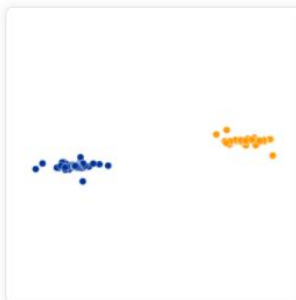
Parameters choice [illustration](#)



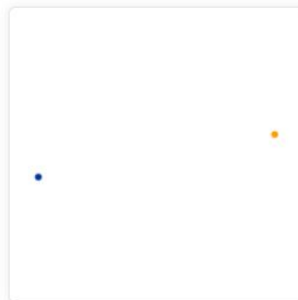
Original



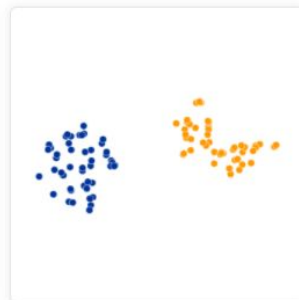
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Step: 10



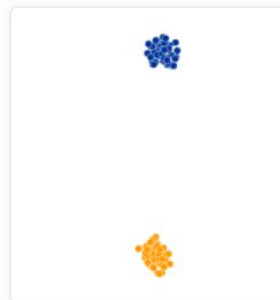
Perplexity: 30
Step: 20



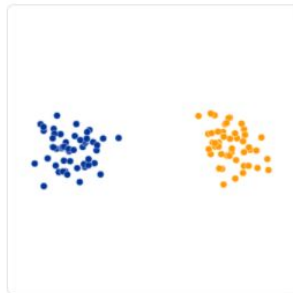
Perplexity: 30
Step: 60



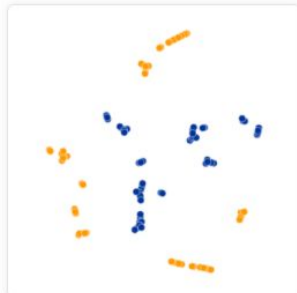
Perplexity: 30
Step: 120



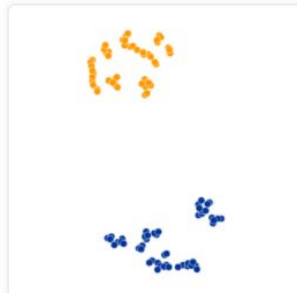
Perplexity: 30
Step: 1,000



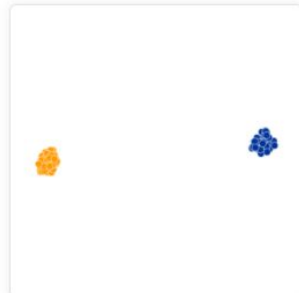
Original



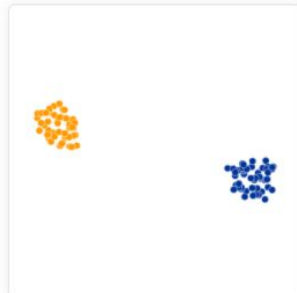
Perplexity: 2
Step: 5,000



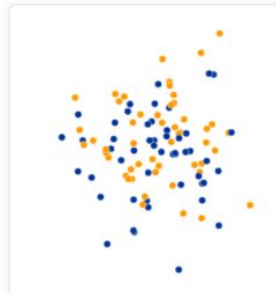
Perplexity: 5
Step: 5,000



Perplexity: 30
Step: 5,000



Perplexity: 50
Step: 5,000



Perplexity: 100
Step: 5,000